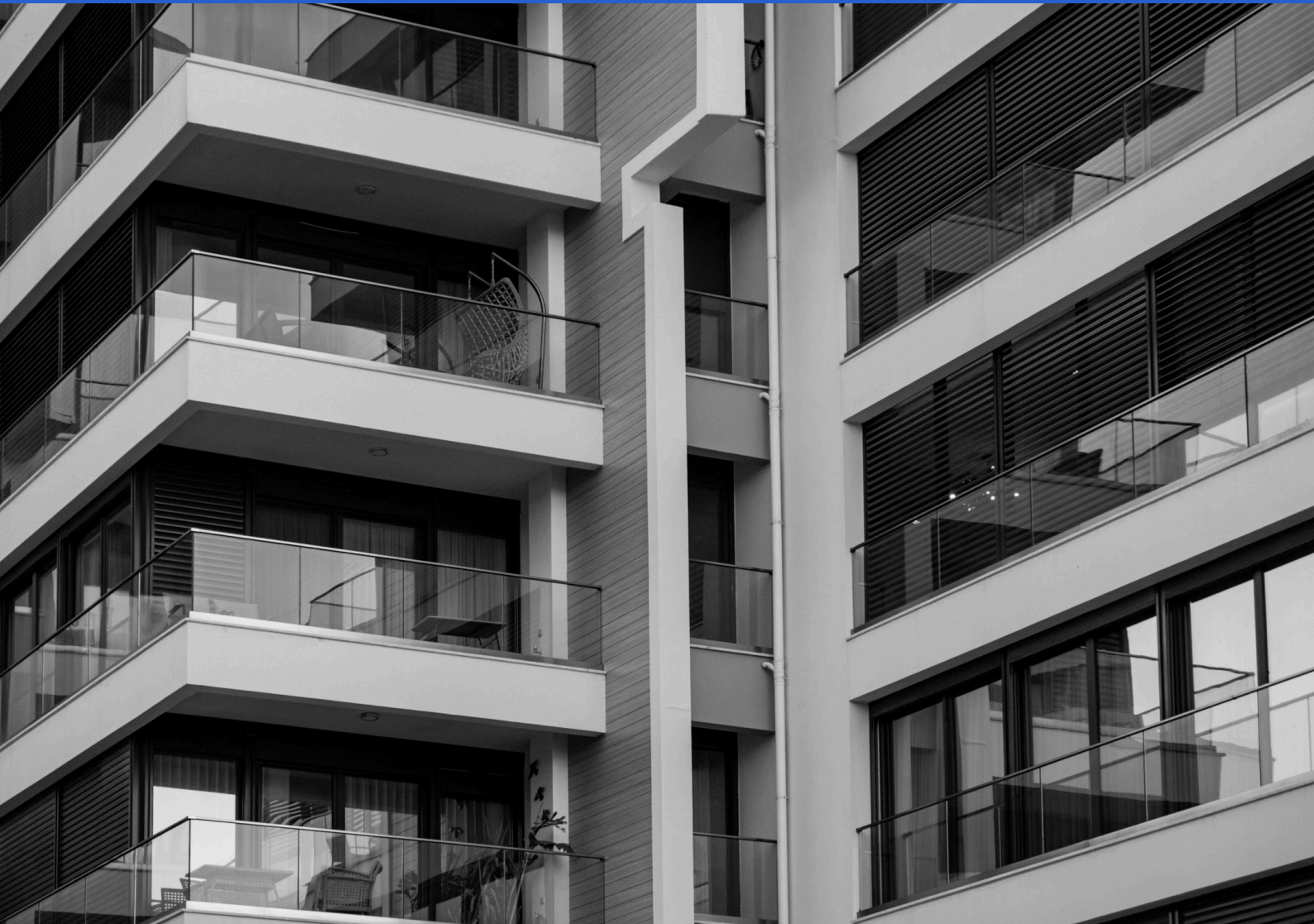


PRESENTED BY:  
**GROUP 8**

SUPERVISOR:  
**MR. THABISO LUNGA APHANE**



# PROJECT PROPOSAL

VALORA

PROPERTY INTELLIGENCE AND PRIVATE  
COMMUNITY MANAGEMENT HUB

# THE PURPOSE:

In South Africa, property buyers and investors face a significant information gap, specifically in housing complexes/private gated communities. Currently, we have platforms like Property24 and Private Property show listings, however these platforms barely provide insight on whether there is potential growth for the area, if the pricing on the property is fair, or what return value an investor can expect. There is also the issue of these housing complexes/private gated communities lacking a unified system to report complaints or disturbances, find trusted local service providers or to track the status of living issues affecting their daily lives within the property.

VALORA is a Private Property Website and Community Management System that aims to combine the listing and viewing of properties within a specific housing complex/private gated community, whilst also giving the residents a private digital platform to voice their concerns. On the property side, for the buyers, investors and real estate agents, VALORA provides AI valuation, growth score, rental yield, as well as a 10-year ROI. On the community living side, relating to the homeowners and residents, VALORA provides a complaint tracker, service provider directory and HOAs.

VALORA, helps to solve a genuine, documented problem in the South African property and community management landscape. It simplifies access to property intelligence whilst also giving the residents a functional, transparent system to improve their life within the housing complexes/private gated communities.

## ROLE ALLOCATION:

| STUDENT NO. | STUDENT NAME             | SPECIALIZATION | ROLE              |
|-------------|--------------------------|----------------|-------------------|
| 402411706   | Mrs. Tabassum Beemath    | SD             | Frontend          |
| 402412302   | Mr. Linda Mhlangu        | SD             | Frontend          |
| 402413832   | Mr. Aadil Mahomed        | ETECH          | Databases         |
| 402412737   | Mr. Dillan Garahushoma   | ETECH          | Backend           |
| 402412770   | Miss. Kaitlin Biggas     | ETECH          | Databases         |
| 402413451   | Mr. Levarshan Gabriel    | ETECH          | Backend           |
| 402413668   | Miss. Aishwariya Chetty  | ETECH          | Charts, Databases |
| 402413754   | Mr. Darshan Premchand    | ITSM           | Documentation     |
| 402412712   | Miss. Kelebongile Tshweu | NET            | Charts, Databases |

- DOCUMENTATION & PRESENTATION DONE BY EVERYONE

# DIFFERENT USERS AND THEIR ROLES:

- Resident – residents would have full community features such as reporting complaints, finding trusted local service providers, real-time alerts and basic property data for their own unit.
- Investor – investors would have full property intelligence which includes a 10-year ROI, valuation retails and rental yield, as well as read-only access to community complaint trends, in order to assess risks.
- Estate Agent – estate agents would be able to list the different options of properties within the housing complex/private gated community, as well as being able to view the living score.
- Casual Visitor – casual visitors would be able to receive a free property valuation estimate, as well as a basic suburb growth score.
- HOA Admin – the HOA admin would be in charge of verifying service providers, moderating complaints, posting announcements and managing AI training data.

# IMPLEMENTATION LANGUAGE:

- Backend: Python with FastAPI (Since there is an AI model being implemented, Python with FastAPI will be used)
- Frontend: JavaScript + React (Since the website will be used on both mobile and desktop devices, React enables reusable components and JavaScript has faster development cycle and learning curve)
- Database: PostgreSQL (This will be used to combine property data, community data and users)
- AI/ML Layer: Python + scikit-learn + XGBoost/FastAPI (This will be used in serving valuation models, as well as complaint categorization)
- Hosting: AWS (This will be used for better speed and scalability)

# THE SDLC MODEL:

The project will be developed using an Agile SDLC with the Scrum framework. The team selected this approach because it enables them to adapt their AI development work through continuous user feedback which helps maintain product market relevance. The project schedule includes 14 bi-weekly sprints, which begin with foundational setup work for authentication and then progress to complex AI features that include valuation and investment metrics, before finishing with security audits and a pilot launch.

### SCRUM ROLES:

- Product Owner: Manages the backlog and prioritizes value.
- The Scrum Master helps the team progress by eliminating all technical and data-related obstacles.
- The Development Team creates the software through its various functional software increments
- Local suburb residents provide essential feedback during review processes as Stakeholders.

The team will utilize Jira/Linear for planning, GitHub for CI/CD, Figma for design, and specialized tools like MLflow for AI model tracking and Sentry for performance monitoring.

Agile methods achieve their main purpose of protecting against scope creep and AI model performance issues and development of unneeded user features.

VALORA aims to democratize property data. The platform combines a service directory with property search and a living score to deliver an all-inclusive representation of neighborhood vitality and property worth. The platform achieves both scalability and user-centered design through its use of a modern tech stack and Scrum framework. The software delivery process produces functional software every two weeks which enables ongoing enhancements until the final product reaches better quality standards..